

Reproduction and Expanded Analysis of LIN28A-Mediated ER-Associated Translational Repression

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Abstract

This report summarizes a six-week project designed to reproduce and extend key analyses from a reference study on LIN28A-mediated regulation in mouse embryonic stem cells. First, gene-level read count data were used to calculate LIN28A CLIP enrichment and ribosome density changes after Lin28a knockdown, allowing the relationship between LIN28A binding and translational de-repression to be examined. Additional localization-based analysis showed that integral membrane-associated genes tended to display greater recovery of ribosome density than nuclear or cytoplasmic genes, and the top 5% of de-repressed genes showed higher LIN28A CLIP enrichment than the remaining genes. Second, ribosome-protected fragment distributions around start codons were reconstructed for both siLuc and siLin28a conditions, revealing a stronger footprint density peak near the translation initiation region after Lin28a knockdown. Third, coverage and Shannon entropy profiles were generated for Mirlet7g, Mirlet7d, and Mirlet7f-1 precursor loci to explore candidate LIN28A binding or crosslinking-induced mutation sites. Together, these analyses are consistent with the model that LIN28A binds let-7 precursor regions and selectively suppresses translation of ER- or membrane-associated mRNAs in mouse embryonic stem cells.

Introduction

LIN28A is known as an RNA-binding protein involved in the regulation of developmental timing, maintenance of stem cell properties, metabolism, and tumor formation [1–3]. Its primary function is to bind to the terminal loop of the let-7 microRNA precursor, thereby inhibiting let-7 maturation by regulating DROSHA/DICER processing and TUTase-mediated uridylation pathways [3–4]. However, the reference study proposed a genome-wide model showing that LIN28A binds not only to pre-let-7 but also to numerous spliced mRNAs, and specifically and selectively inhibits the translation of mRNAs encoding proteins destined for the ER or secretory pathway [5].

The goal of the six-week lab was to reproduce the figures in the paper through actual data processing and visualization steps. Therefore, this report reconstructs the six-week lab around three questions. First, what is the relationship between LIN28A binding affinity and changes in ribosome density following Lin28a knockdown. Second, how does the distribution of ribosome footprints around the translation initiation site change under Lin28a knockdown conditions. Third, how do the error/entropy signals of CLIP reads in the let-7 precursor correlate with their location information on the genome browser. I analyzed the data, generated visualizations, and interpreted the results based on these three questions.

Materials and Methods

The analysis was performed in a Google Colab-based Jupyter notebook, utilizing command-line tools such as samtools, bedtools, and subread/featureCounts via the Bioconda environment. For Python analysis, I used pandas, NumPy, matplotlib, and seaborn. The data used included the CLIP-35L33G BAM, CLIP-let7 series BAM/pileup, RNA-control, RNA-siLuc, RNA-siLin28a, RPF-siLuc, and RPF-siLin28a read counts, as well as GENCODE annotations from the course datapack.

For gene-level analysis, LIN28A CLIP enrichment and ribosome density change were defined as follows.

$$CLIP\ enrichment = \log_2(CLIP_35L33G\ count / RNA_control\ count)$$

$$Ribosome\ density\ change = \log_2\left(\frac{RPF_siLin28a / RNA_siLin28a}{RPF_siLuc / RNA_siLuc}\right)$$

I retained only genes for which the sum of total RNA and RPF counts exceeded 100 and the primary denominator was not zero, excluding those with infinite or missing values. In YourOwnAnalysis 1 (Week 4 lab), I mapped mouse localization annotations to gene IDs to compare the distribution of ribosome density changes across the nucleus, cytoplasm, and integral membrane compartments. I also compared the LIN28A CLIP enrichment of the top 5% of genes based on ribosome density changes with that of the remaining genes using box plots.

For start codon analysis, the `start\_codon` feature was extracted from GTF, and the 5'-end of each read in the RPF BAM file was converted to a position relative to the start codon. The signals were then aggregated around the exon containing the start codon and the range from -50 to +50 nt. In YourOwnAnalysis 2 (Week 5 lab), both the siLuc and siLin28a files were processed in the same manner to directly compare the distributions of the two conditions.

In the CIMS/entropy analysis, Mirlet7g (chr9:106056039-106056126), Mirlet7d (chr13:48689488-48689590), and Mirlet7f-1 (chr13:48691305-48691393) were extracted using samtools view and samtools mpileup. After removing the pileup control character from the base reads string in the pileup, Shannon entropy was calculated based on the base composition at each position. The entropy values were saved in bedGraph format, and for loci corresponding to the negative strand, such as Mirlet7d and Mirlet7f-1, the relative coordinates were calculated as max(pos)-pos to align the 5' to 3' orientation.

Results

LIN28A binding is associated with translational de-repression after Lin28a knockdown

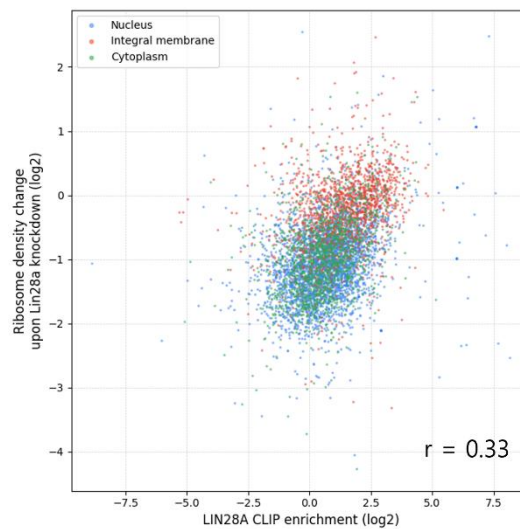


Figure 1. Relationship between LIN28A binding and changes in ribosome density following Lin28a knockdown.

Each point represents a single gene that passed the filtering criteria; the x-axis denotes LIN28A CLIP enrichment (log2), and the y-axis denotes the change in ribosome density following Lin28a knockdown (log2). Point color indicates the protein's subcellular localization annotation: nucleus (blue), integral membrane (red), and cytoplasm (green). Integral membrane genes are distributed toward the relatively higher y-values, and a weak positive correlation ($r = 0.33$) was observed overall. This suggests that transcripts with stronger LIN28A binding may experience greater relief from translational suppression following knockdown.

In Guided Mission 1, when CLIP enrichment and ribosome density changes for all genes were plotted as a scatter plot, the points were clustered in the center, making it difficult to intuitively discern group differences. Figure 1 shows that, after applying a count filter and removing zero-count data, a gentle positive relationship between the two values was observed, and the scatter plot showed a correlation of approximately $r = 0.33$. This aligns with the interpretation of Figure 4D in the original paper, which suggests that the stronger the LIN28A binding to a target, the greater the translational recovery following knockdown.

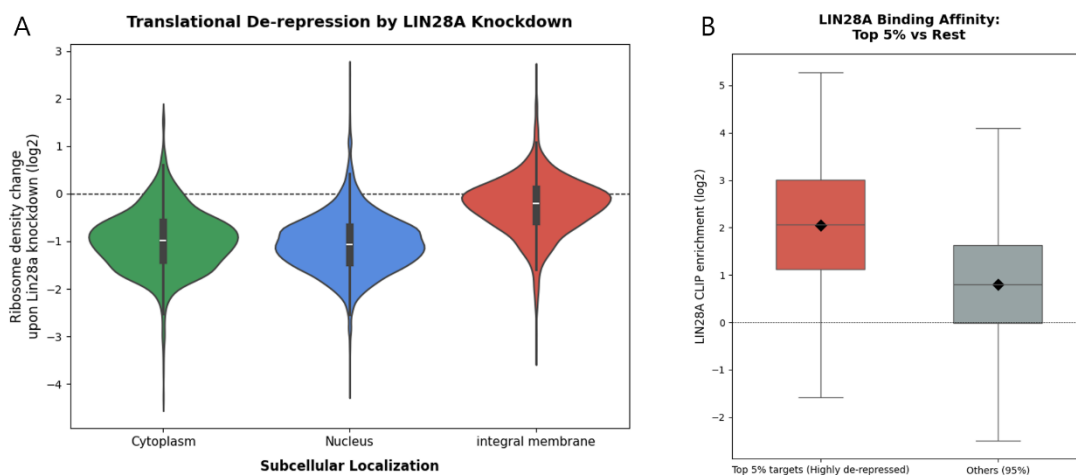


Figure 2. Comparison of translation de-repression following LIN28A knockdown and LIN28A binding strength.

(A) Changes in ribosome density (log2) by cellular location are shown as a violin plot. The integral membrane gene group

exhibited a higher distribution closer to zero compared to the cytoplasm and nucleus gene groups, indicating a tendency for translation suppression to be relatively more relieved following LIN28A knockdown. (B) Compared LIN28A CLIP enrichment ($\log_2$) between the top 5% of genes with the largest ribosome density changes and the remaining 95% of genes. CLIP enrichment was higher in the top 5% of targets, suggesting that transcripts strongly bound to LIN28A are more likely to exhibit a greater release from translational repression following knockdown.

The key improvement of YourOwnAnalysis 1 is that it transforms the issue of overcrowding in scatter plots into a problem of distribution comparison. Figure 2A shows that, in the violin plots by subcellular location, the integral membrane protein group exhibits a distribution where the ribosome density change shifts in a positive direction compared to the nucleus or cytoplasm groups. Furthermore, Figure 2B shows that the top 5% of genes with the largest ribosome density changes exhibit higher average CLIP enrichment than the remaining 95%, thereby reconfirming the relationship between direct binding strength and translational de-repression through a separate group comparison.

Distribution of ribosome footprints around the start codon

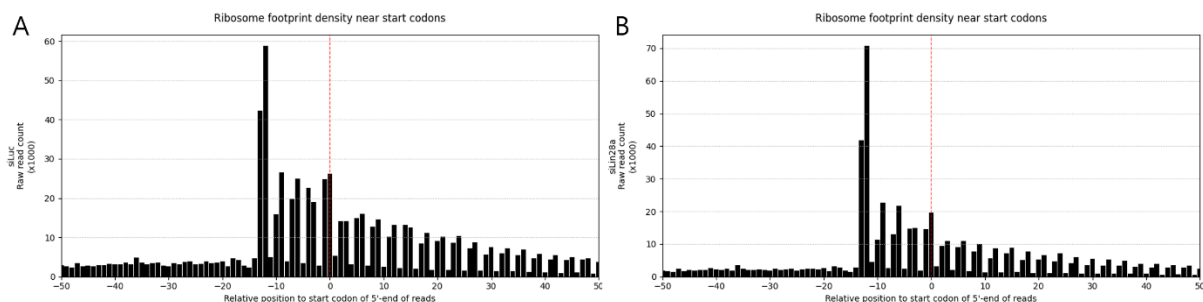


Figure 3. Comparison of ribosome footprint distributions around the start codon.

(A) The 5'-end positions of RPF reads under siLuc control conditions are shown as relative coordinates relative to the start codon. (B) The RPF distribution under siLin28a knockdown conditions is shown in the same manner. The red dashed line indicates the start codon position (0 nt). In both conditions, a distinct peak is observed approximately 10–15 nt upstream of the start codon, demonstrating the typical pattern of ribosome footprint accumulation around the translation initiation site. Under siLin28a conditions, the peak is higher, suggesting that ribosome occupancy around the translation initiation site increased following Lin28a knockdown.

In Guided Mission 2, I analyzed the positions of the 5'-ends of RPF reads relative to the start codon. As described in Figure S5A of the paper, due to the length of the ribosome footprint covering the mRNA, Figure 3A shows a 5'-end peak slightly upstream of the start codon, and the actual bar plot also reveals a distinct peak around –10 nt followed by a distribution in the CDS direction. This demonstrates that the RPF data capture reading events around the translation initiation site.

In YourOwnAnalysis 2, instead of examining only siLuc, the siLin28a condition was added for comparison using the same criteria. The results of the comparison are shown in Figure 3B. As in Figure 3A, a 5'-end peak was observed slightly upstream of the start codon. When using a KDE plot for visual comparison, as shown in Figure 4, the siLin28a condition exhibited a higher density in the region adjacent to the start codon compared to siLuc. This is consistent with the interpretation that target mRNA translation, which was suppressed in the presence of LIN28A,

recovers after Lin28a knockdown, leading to an increase in the ribosome footprint. However, since the KDE is a visualization created by extrapolating raw counts to repeated samples, it should be interpreted as a supplementary figure illustrating distribution differences rather than a quantitative test result.

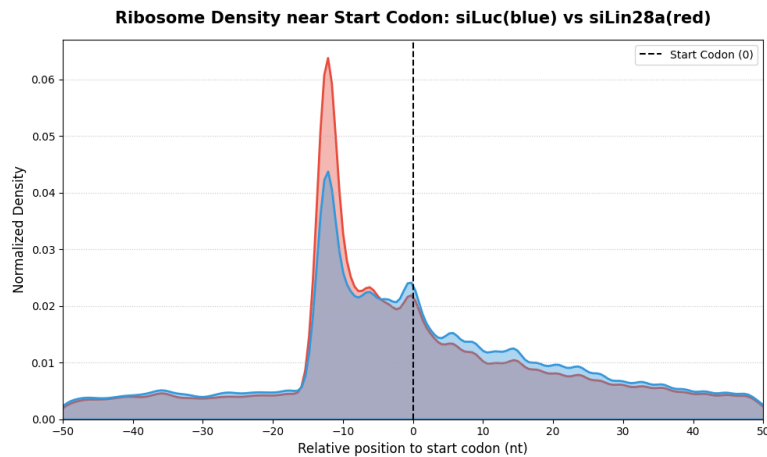


Figure 4. Comparison of RPF density around the start codon under siLuc and siLin28a conditions.

The 5'-end positions of RPF reads were converted to relative coordinates relative to the start codon and plotted using a KDE. Blue represents the siLuc control, red represents the siLin28a knockdown condition, and the black dashed line indicates the start codon position (0 nt). A peak was observed near the start codon, approximately 10–15 nt upstream, in both conditions, with a higher peak observed under the siLin28a condition. This suggests that ribosome density around the translation initiation site increased following Lin28a knockdown.

CIMS/entropy analysis of the let-7 precursor

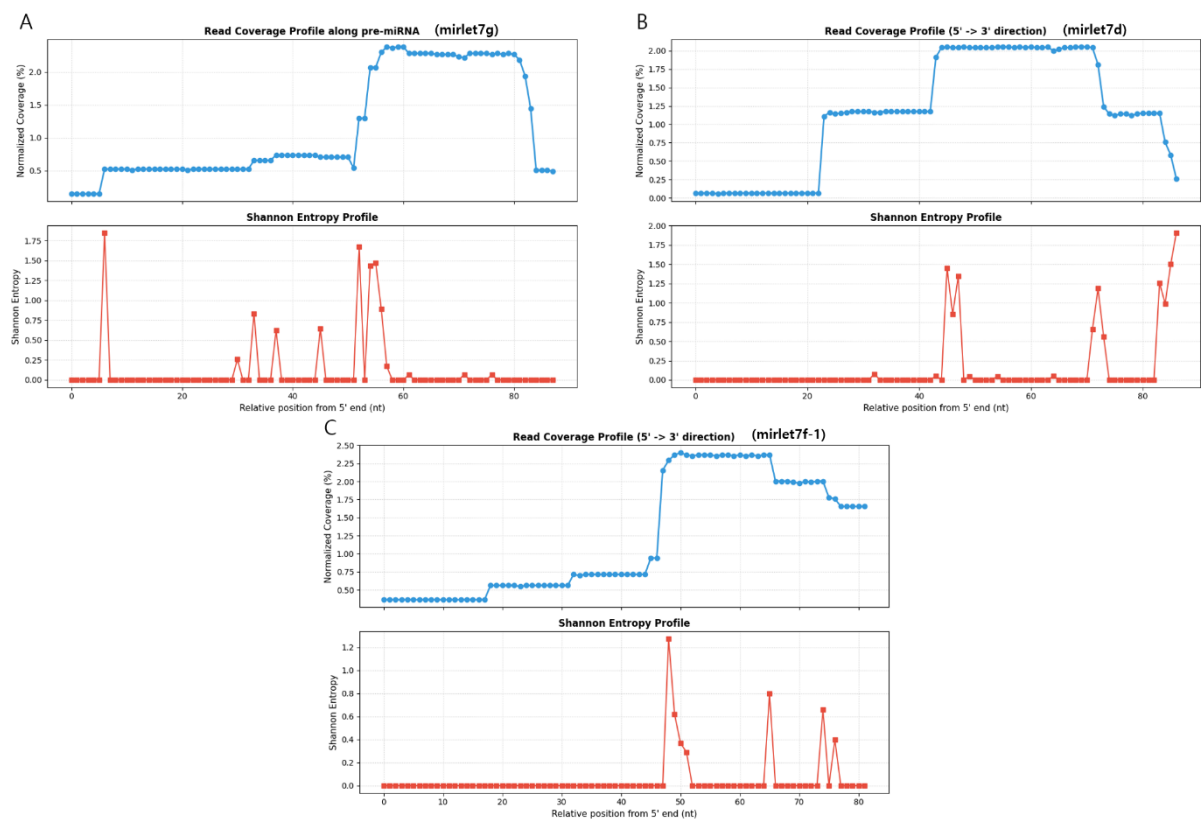


Figure 5. Read coverage and Shannon entropy profiles at let-7 precursor loci.

(A–C) CLIP read coverage and position-specific Shannon entropy are shown for the Mirlet7g, Mirlet7d, and Mirlet7f-1 regions. The top graph shows read coverage normalized along the pre-miRNA, while the bottom graph shows Shannon entropy calculated from pileup base composition. An increase in coverage and a peak in entropy were observed around 50 nt from the 5' end in Mirlet7g and Mirlet7f-1, and around 40 nt from the 5' end in Mirlet7d. This suggests that these positions are likely candidate regions where LIN28A binding or crosslinking-induced mutations are concentrated.

In Mission 3, I calculated the CIMS signal based on the principle that CLIP reads can introduce reverse transcription errors at RNA-binding protein binding sites. By plotting read coverage and Shannon entropy together for each position, I could simultaneously identify locations where reads are densely clustered and where base composition becomes heterogeneous. In YourOwnAnalysis 3, the three precursors—Mirlet7g, Mirlet7d, and Mirlet7f-1—were analyzed separately. Figure 5A, 5C show that for Mirlet7g and Mirlet7f-1, there is a region where coverage or entropy increases sharply around 50 nt from the 5' end, while Figure 5B shows that for Mirlet7d, this region is observed around 40 nt from the 5' end. These results are consistent with the existing model that LIN28A binds to the region surrounding the pre-let-7 terminal loop to inhibit microRNA maturation.

When these three results are integrated, the function of LIN28A can be interpreted at two levels. One is local regulation, in which it directly binds to pre-let-7 to inhibit microRNA maturation; the other is transcriptome-level regulation, in which it suppresses the translation of mRNAs encoding ER or membrane-associated proteins. Mission 1 and YourOwnAnalysis 1 demonstrate this second level at the gene-level, while Mission 2 and YourOwnAnalysis 2 show changes in translation activity following knockdown through the spatial distribution of ribosome footprints. Mission 3 and YourOwnAnalysis 3 reconfirm, at the single-locus level, where LIN28A binds around the let-7 precursor.

Discussion

The strength of this lab is that reproducing the figure was not merely a plotting exercise, but rather an opportunity to experience the entire analytical workflow—from sequencing reads to biological interpretation—in a single session. The featureCounts-based gene-level matrix, RPF 5'-end relative coordinate conversion, pileup-based entropy calculation, and bedGraph and genome browser validation all involve handling data at different resolutions. Nevertheless, these three analyses converged on a single biological model: that LIN28A binding leads to translation repression and let-7 processing inhibition.

The use of a violin plot in YourOwnAnalysis 1 helped compensate for group differences that are easily overlooked in an overcrowded scatter plot. The shift in the distribution of the integral membrane protein group toward the positive y-axis aligns well with the original paper's claim that LIN28A represses ER-associated translation. Furthermore, the comparison showing higher CLIP enrichment in the top 5% of de-repressed targets suggests that binding and translational repression are associated and may be mechanistically linked.

The siLuc/siLin28a KDE comparison in YourOwnAnalysis 2 extends the reproduction of Figure S5A from the original Mission 2. While the mere presence of an RPF peak near the start codon confirms data quality and the resolution of ribosome profiling, placing the knockdown conditions side by side makes the functional interpretation—that footprint density increases in the region adjacent to translation initiation when LIN28A is removed—more clear. However, since the counts were extrapolated to repeated samples to generate the KDE, these results represent a visual comparison of distribution shapes and are limited in their interpretation as statistical significance tests.

The entropy profile in YourOwnAnalysis 3 was useful for intuitively understanding the concept of CIMS. If I simplify the principle that reverse transcriptase errors increase at protein-RNA crosslinking sites using Shannon entropy, I can consider sites with high base composition complexity as binding candidates. However, since the CIMS in the original paper was performed using a more rigorous pipeline that includes replicates, error types, alignment depth, and FDR thresholds, this analysis should be interpreted as limited to the level of visually exploring candidate sites.

If improvements are made in the future, first, the selection of transcript isoforms and the handling of negative-strand transcripts should be applied consistently at every stage. Second, the relationship between CLIP enrichment and ribosome density changes can be tested using permutation tests, or regression models, rather than a simple average comparison. Third, in CIMS analysis, all CIGAR/event information in the pileup string must be carefully interpreted, and background error rates and read depth thresholds must be applied. Fourth, integrating motif prediction and RNA secondary structure information with entropy peaks would allow for a more direct connection between the results of this lab and the AAGNNG/AAGNG motifs and hairpin contexts preferred by LIN28A.

In conclusion, the Week 6 lab began by reproducing a key figure from the LIN28A paper and expanded to include distribution comparisons and locus-level entropy analysis added in this project. The analysis results support the model that LIN28A not only inhibits let-7 precursor processing but also selectively suppresses the translation of ER-associated mRNAs. This is particularly significant because three independent observations—the de-repression of membrane-associated targets, increased RPF around the start codon, and the entropy peak in the let-7 precursor—all provide biological interpretations pointing in the same direction.

References

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